

# Multi-Modal Agentic AI System for Medical Diagnosis & Triage

## Code Review Plan

Code Review Manager - Victor Oluwadare | 20 May 2026 – 24 July 2026 | University of Hertfordshire

### 1. Purpose & Scope

This document defines the code review schedule, per-step checklists, approval gates, and review standards for the project. Reviews are timed to align with Gantt chart milestones so that no phase proceeds until the previous one passes review. I, the Code Review Manager is responsible for conducting or delegating each review, logging defects, and issuing gate approvals.

### 2. Review Standards (apply to every step)

| Standard            | Requirement                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| Code Style          | PEP 8 for Python; ESLint + Prettier for TypeScript. All files must pass linting before review.                               |
| Documentation       | Every function must have a docstring (Python) or JSDoc comment (TypeScript) explaining inputs, outputs, and purpose.         |
| No Hardcoded Values | No model paths, API keys, or thresholds hardcoded. All configurable values in config.py or .env files.                       |
| Test Coverage       | Every new module must have at least one unit test. Critical paths (agent nodes, ONNX inference) require integration tests.   |
| Security            | No credentials in git history. All uploads validated. All DB queries parameterised. Reviewed against OWASP Top 10.           |
| Reproducibility     | RANDOM_SEED=42 used throughout. Model training configs logged to WandB or local file. ONNX exports versioned.                |
| Clinical Safety     | Any change to triage logic or severity thresholds requires two-person sign-off. False-low rate re-tested after every change. |

### 3. Master Review Schedule

Every step has a review deadline 1 day after its Gantt completion date. Steps marked YES in the Gate column are blocking — the next phase cannot begin until the gate is approved by the Code Review Manager.

| Step | Phase | Task                            | Done By | Review By | Review Focus                                                  | Gate?        |
|------|-------|---------------------------------|---------|-----------|---------------------------------------------------------------|--------------|
| 1    | Data  | Dataset Download & Inspection   | 26 May  | 27 May    | Data integrity, EDA notebooks, distribution plots             | No           |
| 2    | Data  | NIH ChestX-ray14 Preprocessing  | 26 May  | 27 May    | Split logic, patient-level leakage prevention, label encoding | No           |
| 3    | Data  | Synthea EHR Feature Engineering | 26 May  | 27 May    | Feature columns, scaler/encoder saved correctly               | No           |
| 4    | Data  | BIO-tag NER Dataset             | 2 Jun   | 3 Jun     | BIO-tag quality, entity coverage, distant supervision logic   | YES — Gate 1 |
| 5    | Data  | Medical Encyclopedia → ChromaDB | 6 Jun   | 7 Jun     | Chunking logic, embedding quality, retrieval spot-check       | No           |

| Step | Phase   | Task                                     | Done By | Review By | Review Focus                                                        | Gate?        |
|------|---------|------------------------------------------|---------|-----------|---------------------------------------------------------------------|--------------|
| 6    | Model 3 | Fine-tune OpenMed NER                    | 16 Jun  | 17 Jun    | Training loop, layer freezing, seqeval F1 $\geq$ 0.82               | YES — Gate 2 |
| 6    | Model 3 | ONNX Export — NER Model                  | 18 Jun  | 19 Jun    | ONNX validation, logit parity, INT8 quantisation, size $\leq$ 250MB | No           |
| 7    | Model 2 | XGBoost Tabular Baseline                 | 17 Jun  | 18 Jun    | Feature importance, GridSearchCV results, cross-val scores          | No           |
| 7    | Model 2 | ClinicalFusion Fusion Network            | 1 Jul   | 2 Jul     | Fusion architecture, multi-task loss weights, ablation table        | YES — Gate 3 |
| 7    | Model 2 | ONNX Export — ClinicalFusion             | 3 Jul   | 4 Jul     | ONNX pipeline validity, accuracy parity, latency < 20ms/record      | No           |
| 8    | Model 1 | ChestVision ResNet-50 / ViT Training     | 8 Jul   | 9 Jul     | Training curves, early stopping, Optuna trial logs, AUC $\geq$ 0.82 | YES — Gate 4 |
| 8    | Model 1 | Grad-CAM & Explainability                | 11 Jul  | 12 Jul    | Heatmap correctness, localisation AUC, anatomical plausibility      | No           |
| 9    | Eval    | Cross-Model Evaluation & Selection       | 12 Jul  | 13 Jul    | Comparison table, model selection justification, metric targets     | YES — Gate 5 |
| 10   | Export  | ONNX Export — All 3 Models               | 14 Jul  | 15 Jul    | All 3 ONNX files valid, latency benchmarks on CPU                   | No           |
| 11   | Export  | INT8 Quantisation                        | 17 Jul  | 18 Jul    | Size limits met (< 250MB each), F1/AUC parity within 0.5 pts        | No           |
| 12   | App     | Next.js + Clerk + NeonDB Scaffold        | 1 Jul   | 2 Jul     | Auth flow, DB schema, route structure, env vars secured             | No           |
| 13   | App     | ONNX Inference API Routes ( $\times 3$ ) | 18 Jul  | 19 Jul    | Request/response schema, error handling, worker threads             | No           |
| 14   | App     | LangGraph Agent (11-node)                | 18 Jul  | 19 Jul    | Node transitions, state schema, MedGemma prompt structure           | YES — Gate 6 |
| 15   | App     | Google Calendar + Gmail API              | 22 Jul  | 22 Jul    | OAuth flow, booking logic, email templates, error fallback          | No           |
| 16   | Prod    | E2E Testing (Playwright / Jest)          | 19 Jul  | 20 Jul    | Synthetic patient scenario coverage, false-low rate < 2%            | No           |

| Step | Phase | Task                      | Done By | Review By | Review Focus                                                | Gate?               |
|------|-------|---------------------------|---------|-----------|-------------------------------------------------------------|---------------------|
| 16   | Prod  | Vercel Deployment + CI/CD | 22 Jul  | 22 Jul    | Live URL, env vars, ONNX model loading, GitHub Actions pass | <b>YES — Gate 7</b> |

#### 4. Phase-by-Phase Review Checklists

The Code Review Manager works through each checklist item and marks it Pass / Fail / N-A. Any Fail blocks the gate for that phase.

| Data Preparation (Steps 1–5) |                                                                                                                                                |                   |                   |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| Review Item                  | What to Check                                                                                                                                  | Result            | Reviewer Sign-off |
| <b>Data Integrity</b>        | Verify no real patient data is present. Confirm NIH CSV row count matches expected 112,120. Confirm Synthea CSVs are synthetic.                | Pass / Fail / N-A |                   |
| <b>Patient-level Split</b>   | Review assign_split() in chestxray_prep.py — confirm split is on Patient ID, not image index. Check no patient appears in both train and test. | Pass / Fail / N-A |                   |
| <b>Label Encoding</b>        | Verify one-hot encoding for all 14 ChestX-ray14 conditions. Confirm 6 target conditions are correctly filtered.                                | Pass / Fail / N-A |                   |
| <b>Feature Engineering</b>   | Review Synthea feature columns: age, BP, oxygen_sat, derived flags (hypoxic, tachycardic). Check SNOMED code mappings.                         | Pass / Fail / N-A |                   |
| <b>Scaler Persistence</b>    | Confirm scaler.pkl and label_encoder.pkl are saved and loadable. Check median imputation for missing values.                                   | Pass / Fail / N-A |                   |
| <b>BIO-tag Quality</b>       | Spot-check 20 random NER samples. Verify B-DISEASE / I-DISEASE / B-ANATOMY / O tags are correctly assigned using SNOMED distant supervision.   | Pass / Fail / N-A |                   |
| <b>ChromaDB Setup</b>        | Query ChromaDB with 3 test terms (pneumonia, cardiac arrest, dyspnoea). Verify top-3 retrieved chunks are medically relevant.                  | Pass / Fail / N-A |                   |

| Model 3 — OpenMed NER (Step 6) |                                                                                                                                |                   |                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| Review Item                    | What to Check                                                                                                                  | Result            | Reviewer Sign-off |
| <b>Layer Freezing</b>          | Confirm encoder layers 0–8 are frozen. Only layers 9–11 and classification head should have requires_grad=True.                | Pass / Fail / N-A |                   |
| <b>Training Config</b>         | Check: AdamW lr=2e-5, warmup_ratio=0.06, weight_decay=0.01, batch=32, epochs=10. Verify CrossEntropy with O-tag class weights. | Pass / Fail / N-A |                   |

|                            |                                                                                                                                    |                   |  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|
| <b>Evaluation Protocol</b> | Confirm segeval span-level F1 (not token-level). Entity F1 $\geq 0.82$ overall; DISEASE recall $\geq 0.85$ (safety-critical).      | Pass / Fail / N-A |  |
| <b>Transfer Gain</b>       | Fine-tuned model must exceed base OpenMed model by $\geq 3$ F1 points on Synthea test set. Document in evaluation report.          | Pass / Fail / N-A |  |
| <b>ONNX Export</b>         | Verify export via HuggingFace Optimum. Check tokeniser is bundled. ONNX entity F1 difference vs PyTorch $< 0.5$ percentage points. | Pass / Fail / N-A |  |
| <b>Size &amp; Latency</b>  | Quantised ONNX (ModernClinical-149M preferred) must be $\leq 150$ MB. CPU inference $< 30$ ms per note.                            | Pass / Fail / N-A |  |

| <b>Model 2 — ClinicalFusion (Step 7)</b> |                                                                                                                                                                                                         |                   |                   |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| Review Item                              | What to Check                                                                                                                                                                                           | Result            | Reviewer Sign-off |
| <b>XGBoost Config</b>                    | Review: 500 estimators, max_depth=6, lr=0.05, subsample=0.8. Confirm GridSearchCV results are logged. Check colsample_bytree=0.8.                                                                       | Pass / Fail / N-A |                   |
| <b>Fusion Architecture</b>               | Verify concatenation: [XGBoost_leaf_64    OpenMed_CLS_768    NER_entity_vector]. Confirm Dense(832+N, 256) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense(256,128). | Pass / Fail / N-A |                   |
| <b>Multi-task Loss</b>                   | Confirm loss = $0.6 \times \text{CrossEntropy}(\text{diagnosis}) + 0.4 \times \text{CrossEntropy}(\text{severity})$ . Verify class weights applied for imbalanced categories.                           | Pass / Fail / N-A |                   |
| <b>Differential LR</b>                   | OpenMed encoder frozen for first 10 epochs, then fine-tuned at lr=5e-6. Main fusion network lr=3e-4. Verify implementation.                                                                             | Pass / Fail / N-A |                   |
| <b>Ablation Study</b>                    | Four variants required: (a) XGBoost-only, (b) NER text-only, (c) Hybrid without fine-tuned NER, (d) Full ClinicalFusion. Macro F1 for each.                                                             | Pass / Fail / N-A |                   |
| <b>Safety Metric</b>                     | False-low rate (critical patients classified as low severity) must be $< 2\%$ on test set. This is the primary safety check.                                                                            | Pass / Fail / N-A |                   |
| <b>ONNX Pipeline</b>                     | Review skl2onnx export for XGBoost + torch.onnx for fusion net. Accuracy parity within 0.01. Severity MAE $< 0.4$ . Latency $< 20$ ms.                                                                  | Pass / Fail / N-A |                   |

| <b>Model 1 — ChestVision (Step 8)</b> |                                                                                                                                                                            |                   |                   |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| Review Item                           | What to Check                                                                                                                                                              | Result            | Reviewer Sign-off |
| <b>Architecture</b>                   | ResNet-50: verify [3,4,6,3] bottleneck blocks, GlobalAvgPool $\rightarrow$ Dropout(0.5) $\rightarrow$ Linear(2048,14) $\rightarrow$ Sigmoid. ViT: 12 blocks, 12-head MHSA. | Pass / Fail / N-A |                   |

|                           |                                                                                                                                                         |                   |  |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|
| <b>Loss Function</b>      | Confirm Asymmetric Loss (ASL) with gamma_neg=4, gamma_pos=0. Verify pos_weight = N_neg/N_pos per class for BCEWithLogitsLoss fallback.                  | Pass / Fail / N-A |  |
| <b>Training Config</b>    | ResNet: AdamW lr=1e-3, cosine decay, batch=64, epochs=100, early-stop patience=10. ViT: lr=1e-4 with 10-epoch warmup.                                   | Pass / Fail / N-A |  |
| <b>Optuna Tuning</b>      | 50 trials, MedianPruner at epoch 20. Search space: lr [1e-4, 5e-3], dropout [0.3,0.6], ASL gamma_neg [2,6]. Log best params.                            | Pass / Fail / N-A |  |
| <b>Augmentation</b>       | Horizontal flip (p=0.5), rotation $\pm 10^\circ$ , brightness/contrast jitter. Confirm no vertical flip (anatomically invalid).                         | Pass / Fail / N-A |  |
| <b>Grad-CAM</b>           | Generate heatmaps for each positive prediction. Overlay on original X-ray. Verify model attends to correct anatomical regions using BBox_List_2017.csv. | Pass / Fail / N-A |  |
| <b>Evaluation Targets</b> | Mean AUC-ROC $\geq 0.82$ across 14 classes. Per-class AUC $\geq 0.75$ . F1 $\geq 0.60$ for top 5 conditions at Youden's J threshold.                    | Pass / Fail / N-A |  |

#### ONNX Export & Quantisation (Steps 10–11)

| Review Item                | What to Check                                                                                                                                                    | Result            | Reviewer Sign-off |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| <b>Export Validity</b>     | All 3 models export without error. Dynamic axes set for batch size. Run 100 test samples through both PyTorch and ONNX; max diff < 1e-4.                         | Pass / Fail / N-A |                   |
| <b>Size Compliance</b>     | ChestVision ResNet-50 $\leq 98$ MB raw, $\leq 25$ MB INT8. ClinicalFusion $\leq 120$ MB. OpenMed NER (149M) $\leq 150$ MB INT8. All must fit Vercel 250MB limit. | Pass / Fail / N-A |                   |
| <b>Latency Benchmarks</b>  | ChestVision < 50ms/image on CPU. ClinicalFusion < 20ms/record. OpenMed NER < 30ms/note. Run onnxruntime speed test and document results.                         | Pass / Fail / N-A |                   |
| <b>Quantisation Parity</b> | INT8 vs FP32: AUC/F1 difference must be < 0.5 percentage points for each model. Reject if degradation exceeds threshold.                                         | Pass / Fail / N-A |                   |

#### Application Build (Steps 12–15)

| Review Item             | What to Check                                                                                                                                                | Result            | Reviewer Sign-off |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| <b>Next.js Scaffold</b> | Review App Router structure, Clerk middleware, NeonDB schema (patients, sessions, diagnoses, appointments, audit_logs tables). Check Drizzle ORM migrations. | Pass / Fail / N-A |                   |

|                              |                                                                                                                                                                                                    |                   |  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|
| <b>Authentication</b>        | Verify Clerk JWT validation on all protected routes. Patient and doctor role separation enforced. No unauthenticated access to /agent or /doctor routes.                                           | Pass / Fail / N-A |  |
| <b>ONNX API Routes</b>       | Each route (/api/models/xray, /api/models/clinical, /api/models/ner) loads model in singleton. Worker threads used for inference. Request validation in place.                                     | Pass / Fail / N-A |  |
| <b>Environment Variables</b> | Confirm .env.local is gitignored. All secrets (Clerk keys, NeonDB URL, HF token, Google OAuth) stored as env vars. No hardcoded credentials in code.                                               | Pass / Fail / N-A |  |
| <b>LangGraph Agent</b>       | Review all 11 nodes: intake → ner → route → xray → clinical → clarify → synthesise → rag → triage → end. Verify state schema TypeScript types. Check MedGemma prompt structure outputs valid JSON. | Pass / Fail / N-A |  |
| <b>Conditional Edges</b>     | Verify routing logic: no X-ray skips xray_node, no EHR skips clinical_node, low confidence triggers clarify_node, SELF_CARE skips triage_node.                                                     | Pass / Fail / N-A |  |
| <b>Google APIs</b>           | OAuth consent screen configured. Calendar slot lookup and booking tested with dummy account. Gmail template reviewed. Error fallback if API unavailable.                                           | Pass / Fail / N-A |  |
| <b>Security</b>              | No SQL injection vectors in NeonDB queries. X-ray uploads validated (file type, size limit). All API routes rate-limited.                                                                          | Pass / Fail / N-A |  |

| <b>Production Deployment (Step 16)</b><br><b>— LAST TECHNICAL STEP</b> |                                                                                                                                                              |                   |                   |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| Review Item                                                            | What to Check                                                                                                                                                | Result            | Reviewer Sign-off |
| <b>E2E Test Coverage</b>                                               | Run full synthetic patient scenario: symptoms → NER → X-ray → ClinicalFusion → MedGemma → RAG → triage → appointment. All nodes must complete without error. | Pass / Fail / N-A |                   |
| <b>False-Low Rate</b>                                                  | Run 50 synthetic critical patients through the agent. Fewer than 1 patient (< 2%) should be classified as low severity. Document results.                    | Pass / Fail / N-A |                   |
| <b>Vercel Deployment</b>                                               | Confirm all 3 ONNX API routes deployed within function size limits. Check cold-start times. Verify environment variables set in Vercel dashboard.            | Pass / Fail / N-A |                   |
| <b>CI/CD Pipeline</b>                                                  | GitHub Actions workflow runs: ONNX validation, unit tests (Jest), E2E tests (Playwright). All checks must pass before merge to main.                         | Pass / Fail / N-A |                   |
| <b>MedGemma Latency</b>                                                | Verify HuggingFace Inference Endpoint responds within acceptable time. Confirm SSE streaming progress indicator works in the UI.                             | Pass / Fail / N-A |                   |
| <b>Audit Logging</b>                                                   | Confirm all LangGraph session states persisted to NeonDB. Verify model                                                                                       | Pass / Fail / N-A |                   |

|                       |                                                                                                                                                 |                   |  |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|
|                       | inputs/outputs logged (excluding raw X-ray images). GDPR compliance check.                                                                      |                   |  |
| <b>Emergency Path</b> | Test EMERGENCY triage path: system instructs patient to call 999, no appointment booked, GP email alert sent. Verify this path is not bookable. | Pass / Fail / N-A |  |

## 5. Approval Gates

A gate is a hard stop. No team member may start the next phase until the Code Review Manager issues written gate approval (email or comment in the project management tool).

| Gate                                | Date               | Pass Criteria                                                                                                         | Unlocks                                           | Approved?              |
|-------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------|
| <b>Gate 1 — Data Ready</b>          | <b>2 Jun 2026</b>  | BIO-tagged NER dataset verified; ChromaDB retrieval spot-checked; all 3 processed datasets confirmed clean            | Proceed to Step 6 (OpenMed NER fine-tuning)       | Yes / No — Date: _____ |
| <b>Gate 2 — NER Model Approved</b>  | <b>18 Jun 2026</b> | sequeval entity F1 $\geq 0.82$ ; transfer gain $\geq 3$ F1 pts; ONNX export validated; size $\leq 250$ MB             | Proceed to Step 7 (ClinicalFusion fusion network) | Yes / No — Date: _____ |
| <b>Gate 3 — ClinicalFusion OK</b>   | <b>2 Jul 2026</b>  | Macro F1 $\geq 0.80$ ; false-low rate $< 2\%$ ; ablation table complete; ONNX latency $< 20$ ms                       | Proceed to Step 9 (cross-model eval)              | Yes / No — Date: _____ |
| <b>Gate 4 — ChestVision OK</b>      | <b>9 Jul 2026</b>  | Mean AUC-ROC $\geq 0.82$ ; per-class AUC $\geq 0.75$ ; Grad-CAM heatmaps reviewed and anatomically plausible          | Proceed to Step 9 (cross-model eval)              | Yes / No — Date: _____ |
| <b>Gate 5 — Models Selected</b>     | <b>13 Jul 2026</b> | Cross-model comparison table complete; final model per task selected and justified; ready for ONNX export             | Proceed to Steps 10–11 (ONNX + quantisation)      | Yes / No — Date: _____ |
| <b>Gate 6 — Agent Approved</b>      | <b>19 Jul 2026</b> | All 11 LangGraph nodes functional; MedGemma returns valid JSON; triage logic verified; Google APIs integrated         | Proceed to Step 16 (E2E testing + deployment)     | Yes / No — Date: _____ |
| <b>Gate 7 — Production Approved</b> | <b>22 Jul 2026</b> | Live Vercel URL working; CI/CD passes; false-low rate $< 2\%$ on E2E test; emergency path verified; audit logs active | Proceed to report writing and submission          | Yes / No — Date: _____ |

## 6. Defect Log Template

The Code Review Manager logs all defects found during review. Critical defects block the gate. Minor defects must be resolved before the next review cycle.

| ID | Step | File / Module | Defect Description | Severity | Status | Assigned To | Resolved By |
|----|------|---------------|--------------------|----------|--------|-------------|-------------|
|    |      |               |                    |          |        |             |             |
|    |      |               |                    |          |        |             |             |

| ID | Step | File / Module | Defect Description | Severity | Status | Assigned To | Resolved By |
|----|------|---------------|--------------------|----------|--------|-------------|-------------|
|    |      |               |                    |          |        |             |             |
|    |      |               |                    |          |        |             |             |
|    |      |               |                    |          |        |             |             |

Severity levels: CRITICAL (blocks gate) | MAJOR (must fix before next review) | MINOR (fix before submission) | INFO (suggestion only)

7. Review Sign-Off

| Role                | Name | Signature & Date |
|---------------------|------|------------------|
| Code Review Manager |      |                  |
| Team Leader         |      |                  |